

Laravel + Angular CRUD

Creating a CRUD application with **Laravel** as the backend API and **Angular** as the frontend provides a clean separation of concerns and a scalable architecture. Below is a step-by-step outline to set up both sides and connect them.

Backend: Laravel API Setup

1. Create Laravel Project

```
composer create-project laravel/laravel api
```

2. Configure Database Edit .env:

```
DB_CONNECTION=mysql
```

```
DB_HOST=127.0.0.1
```

```
DB_PORT=3306
```

```
DB_DATABASE=example
```

```
DB_USERNAME=root
```

```
DB_PASSWORD=
```

3. Create Model & Controller

```
php artisan make:model Product -m
```

```
php artisan make:controller ProductController --api
```

In Product.php:

```
class Product extends Model {  
    protected $fillable = ['name', 'price'];  
    public $timestamps = false;  
}
```

In ProductController.php:

```
public function index() { return Product::all(); }  
public function show($id) { return Product::find($id); }  
public function store(Request $r) { return Product::create($r->all()); }  
public function update(Request $r, $id) { return tap(Product::find($id))->update($r->all()); }  
public function destroy($id) { Product::find($id)->delete(); }
```

4. Define Routes in routes/api.php

```
Route::resource('/products', ProductController::class);
```

Run server:

```
php artisan serve
```

Frontend: Angular Setup

1. Create Angular Project

```
npm install -g @angular/cli@18  
ng new view --routing --style css  
cd view
```

2. Create Service for API Calls

```
ng generate service product
```

In product.service.ts:

```
@Injectable({ providedIn: 'root' })  
export class ProductService {
```

```
baseUrl = 'http://localhost:8000/api/products';

constructor(private http: HttpClient) {}

getAll() { return this.http.get(this.baseUrl); }

get(id: number) { return this.http.get(`${this.baseUrl}/${id}`); }

create(data: any) { return this.http.post(this.baseUrl, data); }

update(id: number, data: any) { return this.http.put(`${this.baseUrl}/${id}`, data); }

delete(id: number) { return this.http.delete(`${this.baseUrl}/${id}`); }

}
```

3. Generate CRUD Components

ng generate component product/index

ng generate component product/create

ng generate component product/edit

ng generate component product/detail

ng generate component product/delete

Implement forms and tables in each component using Angular's **FormsModule** and **HttpClientModule**.

Testing

- Run Laravel API (php artisan serve) on port 8000.
- Run Angular app (ng serve) on port 4200.
- Navigate to <http://localhost:4200> to perform Create, Read, Update, Delete operations.